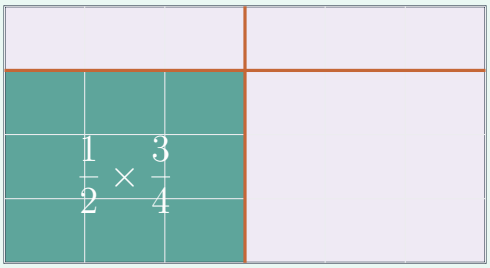
NCERT GANITA PRAKASH • CHAPTER 8

CLASS 7 MATHEMATICS • PART 1

WORKING WITH FRACTIONS

Multiplication • Division • Mixed fractions
Area models • Word problems • Reasoning

An area model makes multiplication visible



The overlap covers $\frac{3}{8}$ of the square

How to study

Every NCERT *Figure it Out* group is solved first, with page numbers so that repeated question numbers are easy to find. Then come illustrated methods, mistake alerts, exam tips and revision questions.

Contents

1 NCERT Figure it Out: complete worked answers **2**
1.1 Set A • Textbook pp. 176–177 (whole numbers times fractions) 2
1.2 Set B • Textbook p. 180 (area models) 3
1.3 Set C • Textbook pp. 183–184 (fraction multiplication in context) 3
1.4 Set D • Textbook pp. 196–198 (division and applications) 4

2 How to understand fractions, not memorise them **5**
2.1 Multiply fractions: pictures and arithmetic 5
2.2 Divide fractions: reciprocal first 6
2.3 Will the answer be bigger or smaller? 6

3 Revision practice and answers **6**
3.1 Challenge corner: five more solved examples 7

NCERT Figure it Out: complete worked answers

Follow the textbook's question groups

This chapter has separate *Figure it Out* sets on textbook pages **176–177, 180, 183–184 and 196–198**. Each set below is numbered independently, matching the book. The picture-based questions are explained with their geometrical or splitting logic.

Set A • Textbook pp. 176–177 (whole numbers times fractions)

A1 • Tenzin's daily milk

Solution. At $\frac{1}{2}$ glass a day, in seven days: $7 \times \frac{1}{2} = \boxed{3\frac{1}{2} \text{ glasses}}$. In January (31 days): $31 \times \frac{1}{2} = \boxed{15\frac{1}{2} \text{ glasses}}$.

Reason. Multiply the daily quantity by the number of days.

A2 • How much canal can be built?

Solution. If 1 km takes 8 days, one day's work is $\boxed{\frac{1}{8} \text{ km}}$. A five-day workweek gives $5 \times \frac{1}{8} = \boxed{\frac{5}{8} \text{ km}}$.

A3 • Oil shared equally among three families

Solution. One family gets $5 \div 3 = \boxed{\frac{5}{3} = 1\frac{2}{3} \text{ L}}$ per week. In four weeks it gets $4 \times \frac{5}{3} = \boxed{\frac{20}{3} = 6\frac{2}{3} \text{ L}}$.

A4 • Moonset, Monday to Thursday

Solution. The Moon sets $\frac{5}{6}$ hour later each day. Monday to Thursday has **three** daily shifts: $3 \times \frac{5}{6} = \frac{15}{6} = 2\frac{1}{2}$ hours. From Monday's 10 pm time, the Thursday moonset is **12:30 am** (half an hour past midnight, early Thursday).

Reason. Count changes between nights, not the number of day names.

A5 (a)–(d) • Multiply and write mixed numbers

(a) $7 \times \frac{3}{5} = \frac{21}{5} = \boxed{4\frac{1}{5}}$,

(b) $4 \times \frac{1}{3} = \frac{4}{3} = \boxed{1\frac{1}{3}}$,

(c) $9 \times \frac{6}{7} = \frac{54}{7} = \boxed{7\frac{5}{7}}$,

(d) $13 \times \frac{6}{11} = \frac{78}{11} = \boxed{7\frac{1}{11}}$.

Check: $7 \times 11 = 77$ with remainder 1, so $78/11 = 7\frac{1}{11}$.

Set B • Textbook p. 180 (area models)

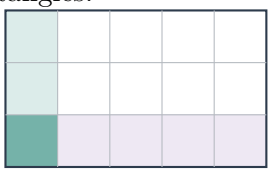
B1 (a)–(e) • Products of unit fractions

(a) $\frac{1}{3} \times \frac{1}{5} = \boxed{\frac{1}{15}}$, (b) $\frac{1}{4} \times \frac{1}{3} = \boxed{\frac{1}{12}}$,

(c) $\frac{1}{5} \times \frac{1}{2} = \boxed{\frac{1}{10}}$, (d) $\frac{1}{6} \times \frac{1}{5} = \boxed{\frac{1}{30}}$,

(e) $\frac{1}{12} \times \frac{1}{18} = \boxed{\frac{1}{216}}$.

Reason. Partition the unit square into rows and columns according to the denominators. The overlap is one of $b \times d$ equal small rectangles.



$\frac{1}{3} \times \frac{1}{5}$

1 shaded out of 15

Set C • Textbook pp. 183–184 (fraction multiplication in context)

C1 (a)–(e) • Water tank fills at 7/10 per hour

Solution. Multiply the hourly fraction by the hours:

(a) $\frac{1}{3}$ h : $\frac{7}{10} \times \frac{1}{3} = \boxed{\frac{7}{30}}$,

(b) $\frac{2}{3}$ h : $\frac{7}{10} \times \frac{2}{3} = \boxed{\frac{7}{15}}$,

(c) $\frac{3}{4}$ h : $\frac{7}{10} \times \frac{3}{4} = \boxed{\frac{21}{40}}$,

(d) $\frac{7}{10}$ h : $\frac{7}{10} \times \frac{7}{10} = \boxed{\frac{49}{100}}$.

(e) To fill one whole tank: $\frac{7}{10}\,t = 1$, so $t = \frac{10}{7} = \boxed{1\frac{3}{7} \text{ hours}}$, assuming a constant rate.

C2 (a)–(c) • Somu divides the remaining land

Road takes $\frac{1}{6}$ of the original land, leaving $1 - \frac{1}{6} = \frac{5}{6}$.

(a) Krishna: $\frac{1}{2} \times \frac{5}{6} = \boxed{\frac{5}{12}}$ of original land.

(b) Bora: $\frac{1}{3} \times \frac{5}{6} = \boxed{\frac{5}{18}}$ of original land.

(c) Somu: $\frac{5}{6} - \frac{5}{12} - \frac{5}{18} = \frac{30-15-10}{36} = \boxed{\frac{5}{36}}$.

Check: $\frac{1}{6} + \frac{5}{12} + \frac{5}{18} + \frac{5}{36} = 1$.

Reason. The phrase *"of the remaining land"* refers to $\frac{5}{6}$, not the original whole.

C3–C5 • Rectangle, saplings and weights

C3: $3\frac{3}{4} \times 9\frac{3}{5} = \frac{15}{4} \times \frac{48}{5} = \boxed{36 \text{ sq ft}}$.

C4: Four saplings make **three gaps**: $3 \times \frac{3}{4} = \boxed{\frac{9}{4} = 2\frac{1}{4} \text{ m}}$.

C5: $\frac{12}{15} \times 500 = 400$ g. But $\frac{3}{20} \times 4$ kg $= \frac{3}{20} \times 4000 = 600$ g. Hence $\boxed{\frac{3}{20} \text{ of } 4 \text{ kg}}$ is heavier by 200 g.

Set D • Textbook pp. 196–198 (division and applications)

D1 • Evaluate every division in the grid

Solution. Keep the first fraction, flip the *divisor*, and multiply.

(a) $3 \div \frac{7}{9} = 3 \times \frac{9}{7} = \boxed{\frac{27}{7}}$,

(b) $\frac{14}{4} \div 2 = \frac{14}{4} \times \frac{1}{2} = \boxed{\frac{7}{4}}$,

(c) $\frac{2}{3} \div \frac{2}{3} = \boxed{1}$,

(d) $\frac{14}{6} \div \frac{7}{3} = \boxed{1}$,

(e) $\frac{4}{3} \div \frac{3}{4} = \frac{4}{3} \times \frac{4}{3} = \boxed{\frac{16}{9}}$,

(f) $\frac{7}{4} \div \frac{1}{7} = \boxed{\frac{49}{4}}$,

(g) $\frac{8}{2} \div \frac{4}{15} = 4 \times \frac{15}{4} = \boxed{15}$,

(h) $\frac{1}{5} \div \frac{1}{9} = \boxed{\frac{9}{5}}$,

(i) $\frac{1}{6} \div \frac{11}{12} = \boxed{\frac{2}{11}}$,

(j) $3\frac{2}{3} \div 1\frac{3}{8} = \frac{11}{3} \div \frac{11}{8} = \boxed{\frac{8}{3}}$.

D2 (a)–(c) • Select the correct expressions

(a) Lace: Each bag uses $\frac{1}{4}$ m. $8 \div \frac{1}{4} = \boxed{32 \text{ bags}}$. This is choice **(iii)**.

(b) Badges: $\frac{1}{2}$ m makes eight badges. $\frac{1}{2} \div 8 = \boxed{\frac{1}{16} \text{ m per badge}}$. This is choice **(iv)**.

(c) Loaves: Five kilograms divided into $\frac{1}{6}$ kg portions: $5 \div \frac{1}{6} = \boxed{30}$. Choice **(iii)**; the expression 5×6 in choice **(iv)** is *also equivalent* to it.

Reason. The word *"per"* or *"how many portions"* helps distinguish multiplication from division.

D3–D5 • Rotis, Sridharacharya's sum and reading

D3: Six rotis require half the flour used for 12: $\frac{1}{4} \div 2 = \boxed{\frac{1}{8} \text{ kg}}$.

D4: $1 \div \frac{1}{6} + 1 \div \frac{1}{10} + 1 \div \frac{1}{13} + 1 \div \frac{1}{9} + 1 \div \frac{1}{2} = 6 + 10 + 13 + 9 + 2 = \boxed{40}$.

D5: Mira reads $\frac{1}{5} + \frac{3}{10} = \frac{1}{2}$ of 400 pages. She has $\frac{1}{2} \times 400 = \boxed{200 \text{ pages}}$ left.

D6–D8 • Petrol, travel and shared cake

D6: $2\frac{3}{4} = \frac{11}{4}$ litres. Distance $= 16 \times \frac{11}{4} = \boxed{44 \text{ km}}$.

D7: Time saved $= 5\frac{1}{6} - \frac{1}{2} = \frac{31}{6} - \frac{3}{6} = \frac{28}{6} = \boxed{4\frac{2}{3} \text{ hours}}$.

D8: Cake left $= 1 - \frac{4}{5} = \frac{1}{5}$. Each of three friends gets $\frac{1}{5} \div 3 = \boxed{\frac{1}{15}}$ of whole cake.

D9 • Compare the product without difficult multiplication

Both $\frac{565}{465} > 1$ and $\frac{707}{676} > 1$. Multiplying either by another number bigger than 1 makes it larger.
Solution. Correct options: $\boxed{\text{(a), (c) and (e)}}$.

D10 • Shaded fraction of the large square

Solution. $\boxed{\frac{3}{32}}$. The shaded piece is a trapezoid within the bottom-right $\frac{1}{4}$ square. Its parallel vertical sides are $\frac{1}{2}$ and $\frac{1}{4}$ of the whole-square side, and their separation is $\frac{1}{4}$. Hence the area fraction is $\frac{1}{2}(\frac{1}{2}+\frac{1}{4}) \times \frac{1}{4} = \frac{3}{32}$.



Shaded: $\frac{3}{32}$

D11 • Ants reach mango tree and sugarcane field

Solution. $\boxed{\text{Mango tree: } \frac{29}{32}}$, $\boxed{\text{Sugarcane: } \frac{3}{32}}$.
Reason. Read the *number of branches* at each red junction (Fig. 8.7). The first two split into halves; the third junction sends equal shares along **four** branches, and the final one into two.
Mango share $= \frac{1}{2}+\frac{1}{4}+\frac{1}{16}+\frac{1}{16}+\frac{1}{32} = \frac{29}{32}$.
Sugarcane share $= \frac{1}{16}+\frac{1}{32} = \frac{3}{32}$.
Check: $\frac{29}{32}+\frac{3}{32} = 1$ (everyone has reached a food source).

D12 • A beautiful telescoping product

Solution. $1-\frac{1}{2} = \boxed{\frac{1}{2}}$; $(1-\frac{1}{2})(1-\frac{1}{3}) = \frac{1}{2}\cdot\frac{2}{3} = \boxed{\frac{1}{3}}$.

$$(1-\tfrac{1}{2})(1-\tfrac{1}{3})(1-\tfrac{1}{4})(1-\tfrac{1}{5}) = \boxed{\tfrac{1}{5}}.$$

$$\prod_{k=2}^{10}\left(1-\frac{1}{k}\right) = \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4}\cdots\frac{9}{10} = \boxed{\frac{1}{10}}.$$

General rule: for an integer $n \geq 2$,

$$\prod_{k=2}^{n}\left(1-\frac{1}{k}\right) = \boxed{\frac{1}{n}}.$$

All intermediate factors cancel between numerators and denominators.

How to understand fractions, not memorise them

Multiply fractions: pictures and arithmetic

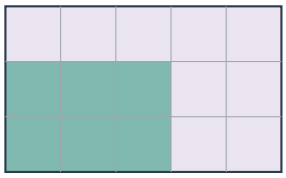
The rule from the textbook

For positive denominators b, d:

$$\boxed{\frac{a}{b}\times\frac{c}{d} = \frac{a\times c}{b\times d}}.$$

Whole number example: $3\times\frac{2}{5} = \frac{6}{5} = 1\frac{1}{5}$.

Two fractions: $\frac{3}{5} \times \frac{2}{3} = \frac{6}{15} = \frac{2}{5}$.
Mixed number: $1\frac{1}{2} \times 2\frac{2}{3} = \frac{3}{2} \times \frac{8}{3} = 4$.



$\frac{3}{5} \times \frac{2}{3}$
6 of 15 shaded
$= \frac{2}{5}$

Cross-cancellation: make the numbers smaller first

$$\frac{14}{15} \times \frac{25}{42} = \frac{1 \times 5}{3 \times 3} = \boxed{\frac{5}{9}}.$$

Cancel common factors **between a numerator and a denominator**, never between two numerators.

Divide fractions: reciprocal first

The flip-and-multiply rule

$$\boxed{\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}}, \qquad c \neq 0.$$

Example: $\frac{2}{3} \div \frac{3}{5} = \frac{2}{3} \times \frac{5}{3} = \frac{10}{9} = 1\frac{1}{9}$.
A whole number is a fraction too: $\frac{1}{4} \div 5 = \frac{1}{4} \times \frac{1}{5} = \frac{1}{20}$.
Sanity check: $6 \div \frac{1}{4} = 24$. A quarter fits in six exactly 24 times.

Will the answer be bigger or smaller?

Action (positive numbers)	What happens?	Example
Multiply by between 0 and 1	Smaller product	$8 \times \frac{1}{4} = 2$
Multiply by greater than 1	Bigger product	$8 \times \frac{3}{2} = 12$
Divide by between 0 and 1	Bigger quotient	$8 \div \frac{1}{4} = 32$
Divide by greater than 1	Smaller quotient	$8 \div 2 = 4$

Common mistakes to avoid

- **Don't add denominators** when multiplying: $\frac{1}{2} \times \frac{1}{3} \neq \frac{2}{5}$; the answer is $\frac{1}{6}$.
- **Don't flip the first fraction** in division. Flip the *divisor only*.
- Change $2\frac{3}{4}$ into $\frac{11}{4}$ *before* multiplying or dividing.
- Four saplings in a row make **three gaps**, not four.
- "One-third of the *remaining* land" is not one-third of the original land.

Revision practice and answers

Try independently before looking at the solutions

1. Evaluate $\frac{5}{6} \times \frac{9}{10}$ and simplify.
2. Evaluate $2\frac{1}{4} \div \frac{3}{4}$.
3. A ribbon $\frac{5}{6}$ m long is divided equally among five cards. How much per card?
4. Decide without full multiplication: is $\frac{7}{9} \times \frac{4}{3}$ greater or smaller than $\frac{7}{9}$?
5. What is $(1 - \frac{1}{2})(1 - \frac{1}{3}) \cdots (1 - \frac{1}{12})$?

Worked answer key

1. $\frac{5}{6} \times \frac{9}{10} = \frac{45}{60} = \boxed{\frac{3}{4}}$.
2. $2\frac{1}{4} \div \frac{3}{4} = \frac{9}{4} \times \frac{4}{3} = \boxed{3}$.
3. $\frac{5}{6} \div 5 = \frac{5}{6} \times \frac{1}{5} = \boxed{\frac{1}{6}\text{ m}}$.
4. **Greater**, since $\frac{4}{3} > 1$ (indeed the product is $\frac{28}{27}$).
5. All middle factors cancel, leaving $\boxed{\frac{1}{12}}$.

Final memory trick

For **multiplication**: numerator × numerator; denominator × denominator. For **division**: **Keep, Change, Flip** — keep the dividend, change ÷ to ×, flip the divisor. Then simplify and check whether the answer's size makes sense.

Challenge corner: five more solved examples

Challenge 1 • A recipe scales up

A recipe needs $\frac{3}{8}$ kg of flour per cake. What is needed for 6 cakes?

Solution. $6 \times \frac{3}{8} = \frac{18}{8} = \boxed{2\frac{1}{4}\text{ kg}}$.

Challenge 2 • Cutting equal ribbon pieces

How many pieces each $\frac{2}{5}$ m can be cut from 4 m of ribbon?

Solution. $4 \div \frac{2}{5} = 4 \times \frac{5}{2} = \boxed{10\text{ pieces}}$.

Challenge 3 • Is the comparison sensible?

A student says $\frac{5}{7} \times \frac{1}{2} = \frac{6}{7}$. Explain the error.

Solution. It is not sensible: multiplying by $\frac{1}{2}$ must make a positive number **smaller**, not larger.

Multiply numerator and denominator: $\frac{5}{7} \times \frac{1}{2} = \boxed{\frac{5}{14}}$.

Challenge 4 • Change a mixed number first

Solution. $1\frac{3}{4} \div \frac{7}{8} = \frac{7}{4} \times \frac{8}{7} = \boxed{2}$.

Challenge 5 • Spot a cancellation pattern

Solution. $\frac{2}{4} \times \frac{4}{6} \times \frac{6}{8} = \frac{2 \cdot 4 \cdot 6}{4 \cdot 6 \cdot 8} = \boxed{\frac{1}{4}}$.

Reason. Cancel 4 and 6 from numerator and denominator, then simplify 2/8.

Study checklist

Before the test, make sure you can (1) convert mixed numbers to improper fractions; (2) multiply and cancel common factors correctly; (3) divide by a non-zero fraction; (4) interpret *of*, *per* and *shared equally*; and (5) explain one fraction problem using a rectangle or number line.